6. Equilibrium Calculations When Simplifications Fail

In the preceding two sections we have used initial concentrations of solutes as though they were equilibrium concentrations when we performed calculations. It is one that works for many situations. Unfortunately, it does not work in all cases, so now that you have, learned the basic approach to solving equilibrium problems, we will examine those conditions under which simplifying approximations do and do not work. We will also study how to solve problems when the approximations cannot be used.

When Simplifying Assumptions Fail

When a weak acid, HA, ionizes in water, its concentration is reduced as the ions form. If we let x represent the amount of acid that ionizes per liter, the equilibrium concentration becomes

$$[HA]_{equilib} = [HA]_{initial} - x$$

For reasons beyond the scope of this text, the accuracy of acid—base equilibrium calculations is limited to about ± 5%, so as long as x does not exceed 3% of [HA] we assume its value is negligible and say that

$$[HA]_{equilib} \approx [HA]_{initial}$$

It is not difficult to show that for x to be less than or equal to ± 5% of $[HA]_{initial}$, $[HA]_{initial}$ must be greater than or equal to 400 times the value of K_a.

$$[HA]_{initial} \geq 400 \cdot K_a$$

For the equilibrium problems in the preceding sections, this condition was fulfilled, so our simplifications were valid. We now want to study what to do $[HA]_{initial} < 400 \cdot K_a$ (i.e., when we cannot justify simplifying the algebra).

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The Quadratic Solution

When the algebraic equation obtained by substituting quantities into the equilibrium law is a quadratic equation, we can use the quadratic formula to obtain the solution. This is illustrated by the following example.

EX.12

Chloroacetic acid, $HC_2H_2O_2C1$, has $K_a = 1.4 \times 10^{-3}$. What is the pH of a 0.01M solution of $HC_2H_2O_2C1$?

Analysis

Before we begin the solution, we check to see whether we can use usual simplifying approximation. We do this by calculating 400 . K_a and comparing the result to the initial concentration of the acid.

$$400 \,.\, K_a = 400(1.4 \times 10^{-3}) = 0.56$$

The initial concentration of $HC_2H_2O_2C1$ is less than 0.56, so we know the simplification does not work.

Solution

Let's begin by writing the chemical equation and the K_a expression,(We know we must use K_a because the only solute is the acid.) $\mathbf{HC_2H_2O_2C1 = H^+ + C_2H_2O_2C1^-}$

$$K_a = \frac{[H^+][C_2H_2O_2Cl^-]}{[HC_2H_2O_2Cl]} = 1.4 \times 10^{-3}$$

The initial concentration of the acid will be reduced by an amount x as it undergoes ionization to form the ions. From this, let's build the concentration table.

	$HC_2H_2O_2C1$	= H^+	+ $C_2H_2O_2C1^-$
Initial concentration(M)	**0.01**	**0.0**	**0.0**
Change in concentration(M)	**-x**	**+x**	**+x**
Equilibrium concentration(M)	**0.01-x**	**x**	**x**

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Substituting equilibrium concentrations into the equilibrium law gives:

$$\frac{(x)(x)}{(0.010 - x)} = 1.4 \times 10^{-3}$$

This time we cannot neglect x. To solve the problem, we first clear fractions by multiplying both sides by (0.01 - x). This gives

$$x^2 = (0.01 - x)1.4 \times 10^{-3}$$

$$x^2 = (1.4 \times 10^{-5}) - (1.4 \times 10^{-3})x$$

The general quadratic equation has the form:

$$ax^2 + bx + c = 0$$

The values of x that make this equation true are related to the coefficients, a, b, and c by the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Rearranging our equation to follow the general form gives:

$$x^2 + (1.4 \times 10^{-3})x - (1.4 \times 10^{-5}) = 0$$

so we make the substitutions a=1, b=1.4×10^{-3}, and c = -1.4×10^{-5}. Entering these into the quadratic formula gives:

$$x = \frac{-1.4 \times 10^{-3} \pm \sqrt{(1.4 \times 10^{-3})^2 - 4(1)(-1.4 \times 10^{-5})}}{2(1)}$$

$$= \frac{-1.4 \times 10^{-3} \pm \sqrt{2.0 \times 10^{-6} + 5.6 \times 10^{-5}}}{2}$$

$$= \frac{-1.4 \times 10^{-3} \pm \sqrt{5.8 \times 10^{-5}}}{2}$$

$$= \frac{-1.4 \times 10^{-3} \pm 7.6 \times 10^{-3}}{2}$$

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Because of the ± sign we obtain two values for x, but as you saw previously, only one of them makes any sense. Here are the two values;

$$x = 3.1 \times 10^{-3}M \quad \text{and} \quad x = -4.5 \times 10^{-3}M$$

We know that x cannot be negative because that would give a negative concentrations for the ions, so we must choose the first value as the correct one. This yields the following equilibrium concentrations.

$$[H^+] = 3.1 \times 10^{-3}M \qquad [C_2H_2O_2Cl^-] = 3.1 \times 10^{-3}M$$

$$[HC_2H_2O_2Cl] = 0.01 - 0.0031 = 0.007\ M$$

Finally, we calculate the pH of the solution. $pH = -\log(3.1 \times 10^{-3}) = 2.51$

Notice that indeed, x is not negligible compared to the initial concentration, so the simplifying approximation would not have been valid.

Solving by Successive Approximations

If you worked your way through the discussion of the use of the quadratic equation, you'll surely be better able to appreciate the method of successive aproximations. It's not only much faster, particularly with a scientific calculator, but just as accurate.

If we were to attempt to use the simplifying approximation in the preceding example, we would obtain the following;

$$\frac{x^2}{0.010} = 1.4 \times 10^{-3}$$

for which we obtain the solution $x = 3.7 \times 10^{-3}$. We will call this our first approximation to a solution to the equation

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$$\frac{(x)(x)}{(0.010 - x)} = 1.4 \times 10^{-3}$$

Let's put $x = 3.7 \times 10^{-3}$ into the term in the denominator, $(0.010 - x)$, and recalculate x. This gives us

$$\frac{x^2}{(0.010 - 0.0037)} = 1.4 \times 10^{-3}$$

Or,

$$x^2 = (0.006) \times 1.4 \times 10^{-3}$$

Taking only the positive root,

$$\mathbf{x = 2.9 \times 10^{-3}}$$

Notice that this value of x is much closer to the value calculated using the quadratic equation (which gave $x = 3.1 \times 10^{-3}$). Now we'll call 2.9×10^{-3} our second approximation, and repeat the process. This gives us:

$$\frac{x^2}{(0.010 - 0.0029)} = 1.4 \times 10^{-3}$$

$$\mathbf{x^2 = (0.007) \times 1.4 \times 10^{-3} \qquad x = 3.1 \times 10^{-3}}$$

This is the identical value obtained using the quadratic equation. Notice also, that the change in x between the two approximations grew smaller. The second approximation gave a smaller correction than the first. Each succeeding approximation differs from the preceding one by smaller and smaller amounts. We stop the calculation when the difference between two approximations is insignificant.

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7. Buffers: The Control of pH

We all know that acids cause things made of metals, like cars or battery terminals, to corrode faster. Chefs know that adding just a little lemon juice to milk makes it curdle. If the pH of your blood were to change from what it should be, within the range of 7.38 to 7.42, either to 7.0 or to 8.0, you would die. Thus, a change in pH can cause chemical reactions, sometimes entirely unwanted. Fortunately, there are ways to protect systems against large changes in pH.

By a careful choice of solutes, a solution can be prepared that will experience no more than a small change in pH even if strong acid or strong base is added or is produced by some reaction. Such mixtures of solutes are called buffers, because they do just that—they buffer the system against a change in pH. The solution itself is said to be buffered or it is described as a buffer solution. There is almost no area of experimental work in chemistry, or in its applications in other fields, where the concept of buffers is not important. Buffers introduce no new concepts to what we have already studied, just a new application.

Components of Buffers

Usually, **a buffer consists of two solutes**. One provides a weak Bronsted acid and the other provides its conjugate base. If the acid is molecular, then the conjugate base is supplied by a soluble salt of the acid.Thus a common buffer system consists of acetic acid plus sodium acetate,with the salt's acetate ion serving as the Bronsted base . Your blood uses carbonic acid (H_2CO_3) , a weak diprotic acid, and the bicarbonate ion, its conjugate Bronsted base,to maintain a remarkably constant pH in the face of the body's production of organic acids by metabolism. Another common buffer consists of the weakly acidic cation, NH_4^+, supplied by a salt like NH_4Cl, and its conjugate base, NH_3.

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One important point about buffers is the distinction between keeping a solution at a particular pH and keeping it neutral—at a pH of 7. Although it is certainly possible to prepare a buffer to work at pH 7, buffers can be made that will work around any pH value throughout the pH scale. One topic we must study, therefore, is how to pick the weak acid and its salt—and their mole ratio— that would make a buffer good for some chosen pH.

Buffer systems do not protect a solution against a tide of strong acid or base, just relatively small amounts. So another topic we have to consider is the capacity of a buffer, the amount of strong acid or base it can absorb before the pH changes a lot.

How a Buffer Works

For simplicity, we will first confine our discussion to the HA/A^- type of buffer system, like the acetic acid—sodium acetate buffer. Let's first see how this system can buffer a solution.

To work, a buffer must be able to neutralize either a strong acid or strong base that is added. This is precisely what the weak base and weak acid components of the buffer do. If we add extra H to the buffer (from a strong acid) the weak conjugate base can react with it as follows.

$$H^+(aq) + A^-(aq) \rightarrow HA(aq)$$

Thus, the added acid changes some of the buffer's Bronsted base, A^-, to its conjugate (weak) acid, HA. This reaction prevents a large buildup of H^+ that would otherwise be caused by the addition of the strong acid.

A similar response occurs when a strong base is added to the buffer. The OH^- from the strong base will neutralize some HA.

$$HA(aq) + OH^-(aq) \rightarrow A^-(aq) + H_2O$$

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Here the added OH^- changes some of the buffer's Bronsted acid, HA, into its conjugate base, A . This prevents a buildup of OH^-, which would otherwise cause a large change in the pH. Thus, one member of a buffer team neutralizes H^+ that might get into the solution, and the other member neutralizes OH^-.

Calculating the pH of a Buffer Solution

The Acetic Acid - Acetate Ion Buffer System

The acetate buffer is a solution (A mixture of $HC_2H_3O_2$ and $C_2H_3O_2^-$ is called the acetate buffer) of both acetic acid and sodium acetate, which provides the acetate ion solution. The $HC_2H_3O_2$ in the buffer neutralizes OH^- as follows:

$$HC_2H_3O_2(aq) + OH^-(aq) \rightarrow C_2H_3O_2^-(aq) + H_2O$$

and its acetate ion neutralizes H^+ as follows:

$$C_2H_3O_2^-(aq) + H^+(aq) \rightarrow HC_2H_3O_2(aq)$$

Let us now use this buffer system to introduce some important simplifications in buffer calculations that are both legitimate and very useful.

EX.13

To study the effects of a weakly acidic medium on the rate of corrosion of a metal alloy, a student prepared a solution by making it both 0.11 M $NaC_2H_3O_2$ and also 0.09 M $HC_2H_3O_2$. What is the pH of this solution?

Solution

We begin with the chemical equation and the expression for K_a.

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$$HC_2H_3O_2 \rightleftharpoons H^+ + C_2H_3O_2^- \qquad K_a = \frac{[H^+][C_2H_3O_2^-]}{[HC_2H_3O_2]} = 1.8 \times 10^{-5}$$

Let's set up the concentration table this time so we can proceed carefully. We will take the initial concentrations of $HC_2H_3O_2$ and $C_2H_3O_2^-$ to be the values given in the problem. There's no H^+ present from a strong acid, so we set this concentration equal to zero. If the initial concentration of H^+ is zero, its concentration must increase on the way to equilibrium, so under H^+ in the change row we enter +x. The other changes follow from that. Here's the completed table.

	$HC_3H_5O_2$	= H^+	+ $C_3H_5O_2^-$
Initial concentration(M)	**0.09**	**0.0**	**0.11**
Change in concentration(M)	**-x**	**+x**	**+x**
Equilibrium concentration(M)	**0.09-x ≈ 0.09**	**x**	**0.11+ x≈ 0.11**

For buffer solutions the quantity x will always be quite small, so it is safe to make the simplifying approximations. What remains, then, is to substitute the final molar concentrations from the table into the K_a, expression.

$$1.8 \times 10^{-5} = \frac{(x)(0.11)}{(0.090)}$$

An equation that can be derived from the K_a expression, and often used by biochemists, is called the Henderson—Hasselbalch equation,

$$pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{conjugate acid}]}$$

For a solution of acetic acid, $HC_2H_3O_2$, which has: $K_a = 1.8 \times 10^{-5}$ and $pK_a = 4.74$, this equation takes the form

$$pH = 4.74 + \log \frac{[C_2H_3O_2^-]}{[HC_2H_3O_2]}$$

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As an exercise, you should consider deriving the Henderson—Hasselbalch equation from the equilibrium expression corresponding to K_a.

Solving for x gives us:

$$x = \frac{(0.090) \times 1.8 \times 10^{-5}}{(0.11)}$$
$$= 1.5 \times 10^{-5}$$

(Notice how small x is compared to the initial concentrations. The simplification was valid.)

Because x equals $[H^+]$,we now have $[H^+]=1.5 \times 10^{-5}$ M. Then we calculate pH:

$$pH = -\log(1.5 \times 10^{-5})$$
$$= 4.82$$

Thus the pH of the buffer is 4.82.

Permissible Simplifications in Buffer Calculations

There are two useful implifications that we can use in working buffer calculations. The first is the one we made in Ex.17. In buffer solutions the initial concentrations of both the acid and its conjugate base are generally large compared to any changes in concentration that occur as a result of the ionization of the acid.

We will be justified in using the initial concentrations of both the weak acid and its conjugate base as though they were equilibrium values.

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There is a further simplification that can be derived from the units for molar concentration, mol L^{-1}. Let's enter these units for the acid and its conjugate base into the mass action expression. For an acid HA,

$$K_a = \frac{[H^+][A^-]}{[HA]} = \frac{[H^+](\text{mol } A^- \text{ L}^{-1})}{(\text{mol } HA \text{ L}^{-1})} = \frac{[H^+](\text{mol } A^-)}{(\text{mol } HA)}$$

Notice that the unit L^{-1} cancels from numerator and denominator. This means that for a given acid—base pair, the $[H^+]$ is determined by the mole ratio of conjugate base to conjugate acid; we don't have to use molar concentradons.

For buffer solutions only, we can use either molar concentrations or moles in the K_a, or K_b) expression to express the amounts of the members of the conjugate acid- base pair (but we must use the same units for each member of the pair).

A further consequence of the relationship derived above is that the pH of a buffer should not change if the buffer is diluted. Dilution does not change the number of moles of the solutes, so the mole ratio remains constant and so does $[H^+]$.

The Ammonia—Aminonium Ion Buffer

A solution of ammonium chloride in aqueous ammonia is a buffer because it contains the weak acid NH_4^+, released hydissociation of the NH_4Cl, and its conjugate base, NH_3. The buffer is able to neutralize OH^- by the following reaction.

$$NH_4^+ (aq) + OH^- (aq) \rightarrow NH_3(aq) + H_2O$$

The conjugate base can neutralize H^+.

$$NH_3(aq) + H^+ (aq) \longrightarrow NH_4^+ (aq)$$

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EX.14

To study the influence of an alkaline medium on the rate of a reaction, a student prepared a buffer solution by dissolving 0.12 mol of NH_3 and 0.095 mol of NH_4Cl in water. What is the pH of the buffer?

Solution

We begin with the chemical equation and the K_b expression.

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^- \qquad K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = 1.8 \times 10^{-5}$$

The solution contains 0.12 mol of NH_3 and 0.095 mol of NH_4^+ from the complete dissociation of the salt NH_4Cl. We can enter these quantities into the mass action expression and solve for $[OH^-]$.

$$1.8 \times 10^{-5} = \frac{(0.095)[OH^-]}{0.12}$$

Solving for [OH] gives: $[OH^-] = 2.3 \times 10^{-5}$

To calculate the pH, we obtain pOH and subtract it from 14.00.

$$pOH = -\log(2.3 \times 10^{-5}) = 4.64 \qquad pH = 14 - 4.64 = 9.36$$

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